

# Relation Algebra

## §12. Combinatorial functions

(12a)

| definition                                                                                                              | type                                  | note                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $[A] ::= \text{nil} \mid \text{cons}(A, [A])$                                                                           | $\mathcal{A} \rightarrow \mathcal{A}$ | The list type, under the short name it keeps.                                                                                                                |
| $\text{list}(R) = ([\text{nil}, (R \times 1) \text{cons}])$                                                             | $[A] \rightarrow [B]$                 | The relator's action on $R : A \rightarrow B$ : one $R$ per element, the shape untouched.                                                                    |
| $\text{subseq} = ([\text{nil}, \text{cons } u \pi_2])$                                                                  | $[A] \rightarrow [A]$                 | $xs \text{ subseq } ys$ : $ys$ is $xs$ with elements dropped — $\text{cons}$ keeps the head, $\pi_2$ drops it.                                               |
| $\text{prefix} = ([\text{nil}, \text{nil } u \text{cons}])$<br>$\text{prefix} = \text{cat} \circ \pi_1 = \text{init}^*$ | $[A] \rightarrow [A]$                 | $ys$ is an initial segment of $xs$ ; the first $\text{nil}$ is where it stops early. $\text{init} \triangleq \text{snoc} \circ \pi_1$ .                      |
| $\text{suffix} = \text{cat} \circ \pi_2 = \text{tail}^*$                                                                | $[A] \rightarrow [A]$                 | The dual, $\text{tail} \triangleq \text{cons} \circ \pi_2$ ; as a reduce it needs $\text{snoc}$ -lists.                                                      |
| $\text{segment} = \text{suffix } \text{prefix}$                                                                         | $[A] \rightarrow [A]$                 | A contiguous stretch of $xs$ : a suffix, then a prefix of that.                                                                                              |
| $\text{partition} = \text{concat}^\circ$                                                                                | $[A] \rightarrow [[A]]$               | This $\text{cat}$ is restricted to $[A]^+ \times [A] \rightarrow [A]$ , so $ys$ is a list of non-empty segments of $xs$ .                                    |
| $\text{concat} = ([\text{nil}, \text{cat}])$                                                                            | $[[A]] \rightarrow [A]$               | Joins the segments back up, which is why its converse splits a list.                                                                                         |
| $\text{inits}$                                                                                                          | $[A] \rightarrow [[A]]$               | Implements $\frac{\text{prefix}}{\ni}$ , listing the prefixes by increasing length.                                                                          |
| $\text{tails}$                                                                                                          | $[A] \rightarrow [[A]]$               | Implements $\frac{\text{suffix}}{\ni}$ by decreasing length — the opposite order.                                                                            |
| $\text{filter}(p) \triangleq \frac{\text{subseq } \text{list}(p)}{\ni} \text{est}(R^\circ)$                             | $[A] \rightarrow [A]$                 | The longest subsequence of $xs$ whose every element passes $p$ . $\text{est}(R^\circ)$ is $\triangleq \text{est}$                                            |
| $R \triangleq \text{length} \leq \text{length}^\circ$                                                                   | $[A] \rightarrow [A]$                 | The preorder $\text{filter}$ and $\text{takewhile}$ maximise over: the longer list wins. $\geq \triangleq \leq^\circ$                                        |
| $\text{takewhile}(p) \triangleq \frac{\text{prefix } \text{list}(p)}{\ni} \text{est}(R^\circ)$                          | $[A] \rightarrow [A]$                 | The same with $\text{prefix}$ for $\text{subseq}$ : the longest prefix whose every element passes $p$ .                                                      |
| $\text{mss} \triangleq \frac{\text{segment } \text{sum}}{\ni} \text{est}(\geq)$                                         | $[A] \rightarrow A$                   | Maximum segment sum. $\text{segment} = \text{suffix } \text{prefix}$ splits it into $\frac{\text{prefix } \text{sum}}{\ni} \text{est}(\geq)$ on each suffix. |

### §12.1. $\frac{\text{subseq}}{\ni} = ([\text{nil } \frac{1}{\ni}, (\frac{1 \times \ni}{\ni} \text{E}(\text{cons}), \pi_2) \text{cup}])$

(12.1a)

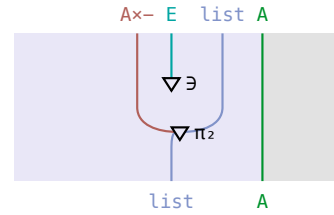
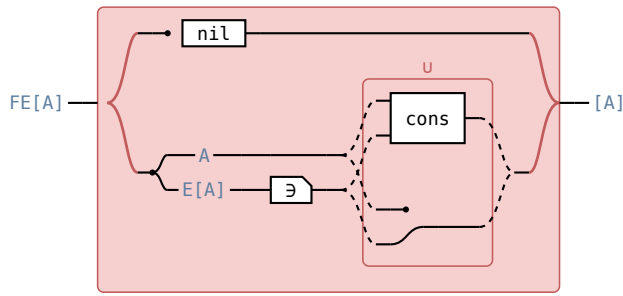
$\text{cup} \triangleq \frac{\pi_1 \ni \cup \pi_2 \ni}{\ni} : EA \times EA \rightarrow EA$ , so  $\frac{R \cup S}{\ni} = (\frac{R}{\ni}, \frac{S}{\ni}) \text{cup}$ .

$$\frac{F(\exists)[\text{nil}, \text{cons } u \pi_2]}{\exists} = [\text{nil } \frac{1}{\exists}, \frac{(1 \times \exists)(\text{cons } u \pi_2)}{\exists}] \quad (12.1b)$$

the set of lists the algebra builds is, from nothing, just **nil**, and from a head and a set of tails, every tail in the set with the head put on or left off — cata-map-calc at **subseq** =  $([\text{nil}, \text{cons } u \pi_2])^\exists$ ,  $\triangleq$  **subseq**.

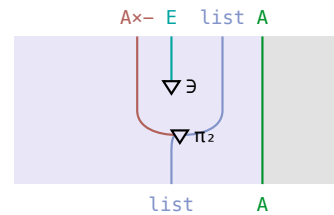
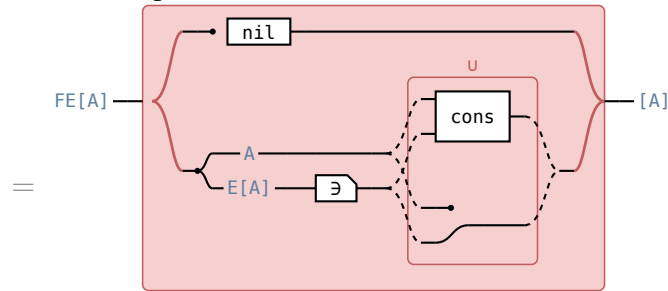
**circuit** — the fork is  $F([A]) = 1 + A \times [A]$ : **nil** above, the pair below

**Hinze–Marsden**



$$\frac{F(\exists)[\text{nil}, \text{cons } u \pi_2]}{\exists}$$

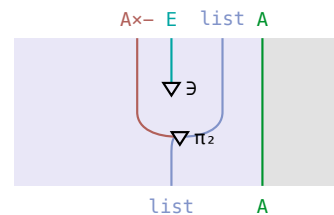
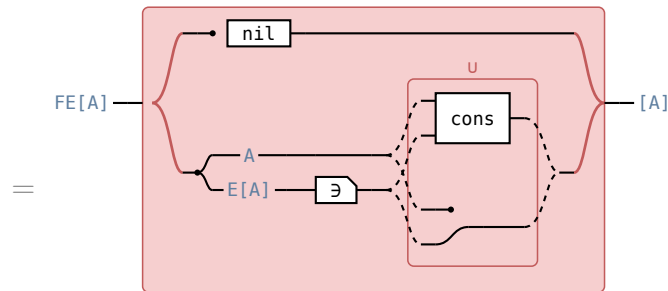
the  $\pi_2$  operand of  $\text{cons } u \pi_2$  under the  $1 \times \exists$  summand of  $F(\exists)$ , i.e.  $(1 \times \exists)\pi_2$



$$\frac{(1+1 \times \exists)[\text{nil}, \text{cons } u \pi_2]}{\exists}$$

$$F(X) = 1 + A \times X \triangleq \text{subseq}$$

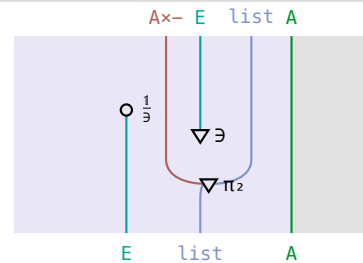
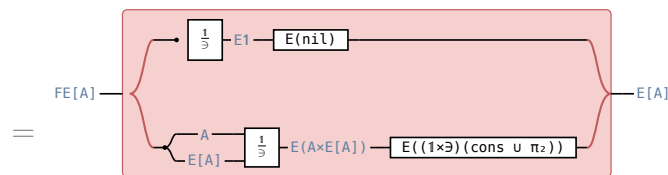
the same operand under  $1 + 1 \times \exists$ , whose  $1 \times \exists$  summand it sits in



$$\frac{[\text{nil}, (1 \times \exists)(\text{cons } u \pi_2)]}{\exists}$$

$$R + S \triangleq [Rl, Sr], l[R, S] = R, r[R, S] = S \text{ — coprod-laws}$$

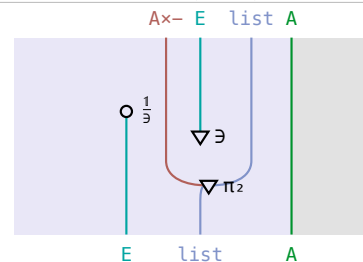
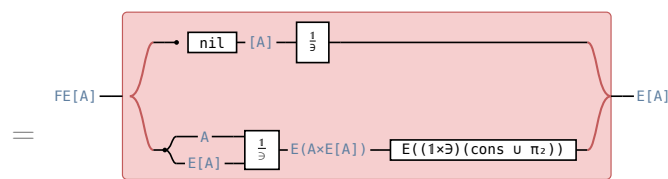
the  $\pi_2$  operand of the second arm  $(1 \times \exists)(\text{cons } u \pi_2)$



$$\frac{[\frac{\text{nil}}{\exists}, \frac{(1 \times \exists)(\text{cons } u \pi_2)}{\exists}]}{\exists}$$

$$\text{coprod-calc at } T := [\text{nil}, (1 \times \exists)(\text{cons } u \pi_2)]$$

the  $\pi_2$  operand under its  $1 \times \exists$ , the arm (12.1c)'s square rewrites,  $(1 \times \exists)\pi_2 = \pi_2 \exists$



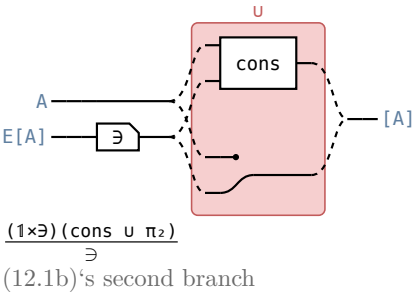
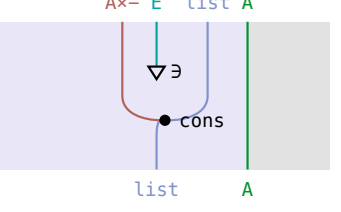
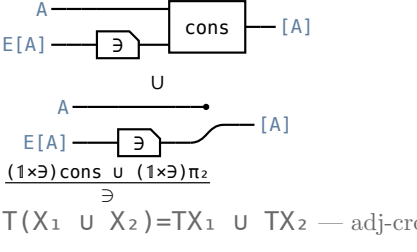
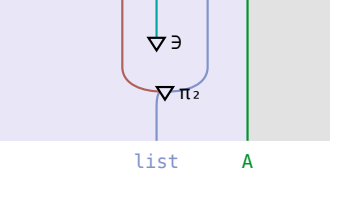
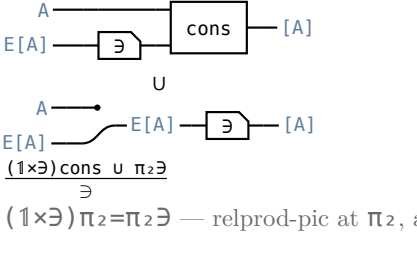
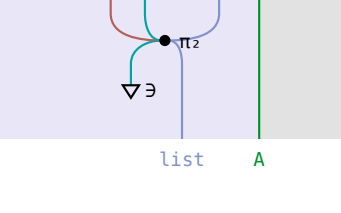
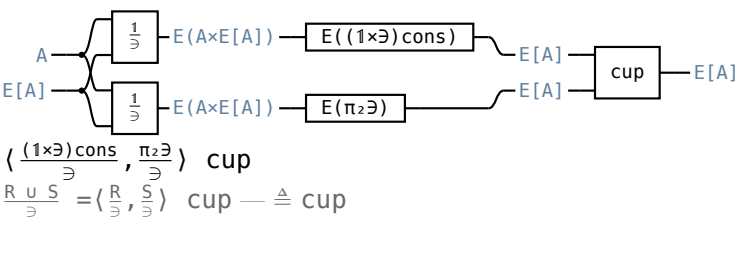
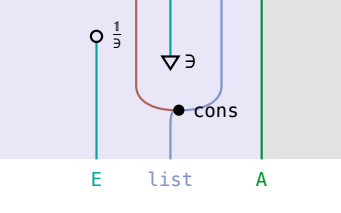
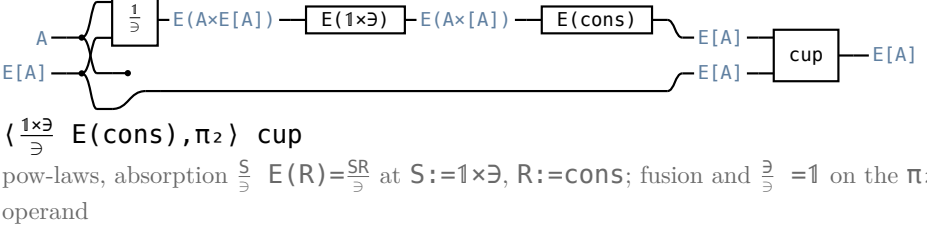
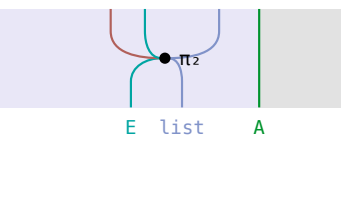
$$[\text{nil } \frac{1}{\exists}, \frac{(1 \times \exists)(\text{cons } u \pi_2)}{\exists}]$$

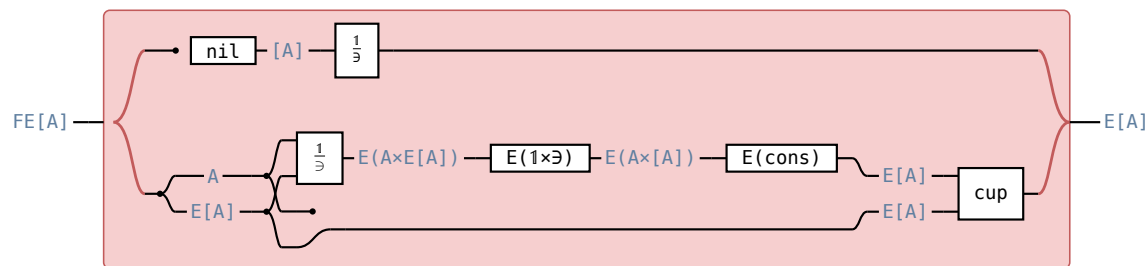
$$\text{pow-laws, } \frac{f}{\exists} = f \frac{1}{\exists} \text{ for } f \text{ a map, at } f := \text{nil}$$

the same operand; the two rows differ only in the **nil** arm

$$\begin{array}{ccc} A \times E[A] & \xrightarrow{\pi_2} & E[A] \\ \downarrow 1 \times \exists & & \downarrow \exists \\ A \times [A] & \xrightarrow{\pi_2} & [A] \end{array}$$

(12.1c)

| $\frac{(1 \times \exists)(\text{cons} \cup \pi_2)}{\exists} = \left( \frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \right) \text{cup}$ <p>power transpose of join: the power transpose of the join of two relations is <math>\left( \frac{R}{\exists}, \frac{S}{\exists} \right) \text{cup}</math>, where <b>cup</b> is the function that returns the union of two sets</p>                                                    | <p>(12.1d)</p>                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                 | Hinze–Marsden                                                                                                                                                                                        |
|  <p><math>\frac{(1 \times \exists)(\text{cons} \cup \pi_2)}{\exists}</math><br/>(12.1b)'s second branch</p>                                                                                                                                                                                                                                            |  <p>the <b>cons</b> operand of <math>\text{cons} \cup \pi_2</math></p>                                            |
|  <p><math>\frac{(1 \times \exists)\text{cons} \cup (1 \times \exists)\pi_2}{\exists}</math><br/><math>T(X_1 \cup X_2) = TX_1 \cup TX_2</math> — adj-cross</p>                                                                                                                                                                                          |  <p>the <math>\pi_2</math> operand of <math>(1 \times \exists)\text{cons} \cup (1 \times \exists)\pi_2</math></p> |
|  <p><math>\frac{(1 \times \exists)\text{cons} \cup \pi_2 \exists}{\exists}</math><br/><math>(1 \times \exists)\pi_2 = \pi_2 \exists</math> — relprod-pic at <math>\pi_2</math>, an equality because <math>1</math> is entire</p>                                                                                                                      |  <p>the <math>\pi_2</math> operand of <math>(1 \times \exists)\text{cons} \cup \pi_2 \exists</math></p>          |
|  <p><math>\frac{(1 \times \exists)\text{cons}}{\exists} \cup \frac{\pi_2 \exists}{\exists} \text{cup}</math><br/><math>\frac{R \cup S}{\exists} = \left( \frac{R}{\exists}, \frac{S}{\exists} \right) \text{cup}</math> — <math>\triangle</math> cup</p>                                                                                             |  <p>the <b>cons</b> operand under its <math>1\%</math></p>                                                      |
|  <p><math>\left( \frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \right) \text{cup}</math><br/>pow-laws, absorption <math>\frac{S}{\exists} E(R) = \frac{SR}{\exists}</math> at <math>S := 1 \times \exists</math>, <math>R := \text{cons}</math>; fusion and <math>\frac{\exists}{\exists} = 1</math> on the <math>\pi_2</math> operand</p> |  <p>the <math>\pi_2</math> operand, bare <math>\pi_2</math></p>                                                 |



$$\left[ \frac{\text{nil}}{\exists}, \left( \frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \right) \text{cup} \right]$$

which writes  $P\text{cons}$ ;  $\text{cons}$  is a map, and there  $P(\text{cons}) = E(\text{cons})$  — powrel-laws.